



General and Supportive Care

Expert consensus guidelines for the prophylaxis and management of tumor lysis syndrome in the United States: Results of a modified Delphi panel

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ABSTRACT

Introduction: Tumor lysis syndrome (TLS), which occurs spontaneously or in response to anticancer treatment, results in the release of intracellular potassium, phosphorus, and nucleic acids into the bloodstream, which results in secondary clinical complications that may be fatal. Prior TLS guidelines do not take into consideration potent novel oncologic agents or contemporary treatment paradigms with increased risk of TLS. Thus, a modified Delphi panel of experts was convened to provide an update for TLS management guidelines based upon a combination of supporting literature and practice consensus.

Methods: A three-round modified Delphi process was implemented. For round 1, nine expert panelists completed a web-based questionnaire developed using published literature. In round 2, panelists were asked to reconsider their answers to questions that did not reach consensus (defined as $\geq 66\%$ agreement among voting panelists). Round 3 was an unblinded, moderated virtual meeting to discuss any remaining questions that did not reach consensus.

Results: Detailed recommendations are given for prophylaxis, monitoring, and management of TLS risks and complications, with hydration being a key element of TLS prophylaxis and management. Guidelines for the management of acute effects of TLS and prevention of long-term renal effects include management of hyperkalemia, hypocalcemia, hyperphosphatemia, and hyperuricemia.

Discussion: Although the control of uric acid levels is quite effective with currently available agents, panelists emphasize the importance of monitoring and treating other dangerous electrolyte abnormalities such as hyperkalemia and hyperphosphatemia. Guidelines from this modified Delphi panel should aid clinicians in preventing and managing TLS.

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Introduction

Tumor lysis syndrome (TLS) can develop when oncologic therapies cause massive cellular destruction, resulting in large amounts of intracellular potassium, phosphorus, and nucleic acids released into the bloodstream [1]. TLS occurs when the accumulation of these cellular contents and electrolytes exceeds the body's homeostatic clearance mechanisms and is usually observed within 12–72 h after initiation of chemotherapy, but it can also occur spontaneously [1–3]. Released adenine and guanine are metabolized into hypoxanthine and xanthine, then by xanthine oxidase into uric acid [1]. At high levels, uric acid and/or calcium phosphate can co-precipitate in kidney tubules, leading to acute kidney injury (AKI) [1,4]. Uric acid can also induce AKI by impairing autoregulation, resulting in decreased renal blood flow, and by causing inflammation, leading to enhanced renal vasoconstriction and oxidative damage [5]. Tumor lysis can also result in the release of cytokines (e.g., tumor necrosis factor, interleukin-6) that cause systemic inflammatory response syndrome and multi-organ failure [6,7].

TLS is generally classified by laboratory and/or clinical criteria. Laboratory TLS is characterized by hyperkalemia, hyperphosphatemia, hyperuricemia, and potentially secondary hypocalcemia [1,2,8]. General clinical symptoms of TLS may include nausea, vomiting, shortness of breath, irregular heartbeat, reduced urine output, lethargy, and/or joint discomfort [9]. Clinical TLS can manifest as cardiac arrhythmia, hypotension, heart failure, increased serum creatinine levels, oliguria, muscle cramps, tetany, seizure, syncope, or sudden death [1,2,4,9]. TLS can be managed prophylactically or therapeutically by closely monitoring and correcting electrolyte imbalances, dehydration, AKI, cardiac arrhythmias, and uric acid levels and by maintaining optimal urine output.

Prior TLS guidelines were developed in an era when traditional cytotoxic chemotherapy approaches were used before the availability of contemporary potent targeted anticancer agents and immunotherapeutics that increase risk of TLS. The National Comprehensive Cancer Network maintains guidelines for various malignancies, but their levels of detail vary for TLS risk, prophylaxis, monitoring, and management [9–15]. Some institutions have developed their own TLS guidelines, which vary widely. For example, there is marked variability in rasburicase use among children's hospitals in the United States [16]. Given the approval of many new targeted, immune-based, and cellular anticancer therapies over the last two decades, there is a need to revisit TLS guidelines and modify them in the context of these new therapies.

The Delphi method is a multistage, systematic process to obtain consensus on a subject from an expert panel [17,18]. To evaluate current clinical practice in the context of older guidelines, TLS experts participated in a modified Delphi panel to define clinically relevant and widely

applicable consensus strategies for the optimization of TLS management.

Materials and methods

The modified Delphi process

A three-round modified Delphi process was employed (Fig. 1). Published literature was used to develop the questionnaire, followed by a live meeting where panelists discussed points on which consensus was not reached in earlier rounds [17,18]. In round 1, panelists anonymously completed a web-based questionnaire. In round 2, panelists were asked to reconsider their answers to questions that did not reach consensus from round 1. Round 3 was an unblinded virtual meeting to discuss the remaining questions in an attempt to reach consensus. This meeting was moderated by the non-panelist author (SCH) to ensure equal participation and feedback on the questions.

Participants

Panelists were US healthcare providers experienced in the management of TLS in varying settings. These expert panelists, some of whom published prior guidelines, had relevant knowledge and experience in TLS management and were willing to participate in the entire process. Panelists provided written informed consent. Questionnaire administration, data acquisition, and analyses were conducted by inScience Communications, Springer Healthcare (New York, NY).

Questionnaire development and administration

Questionnaires were developed based on several existing guidelines that address patient TLS risk assessment, monitoring, prophylaxis, and treatments [2–4,19], with the non-panelist author providing advice. Round 1 questionnaires were completed from October 26 to November 15, 2021, via an online survey platform, individualized round 2 questionnaires were completed from November 29 to December 13, 2021, and the round 3 virtual meeting was held on March 1, 2022. All nine panelists completed both rounds of questionnaires, with eight of the nine panelists present for the entire virtual meeting; one panelist joined for the second half of the meeting. The full questionnaire is available in Supplemental Data.

Analysis

Answers were evaluated for percent agreement for all three rounds. Panelists were not required to respond to all questions. The pre-specified consensus threshold was $\geq 66\%$ agreement among voting panelists for each question [17].

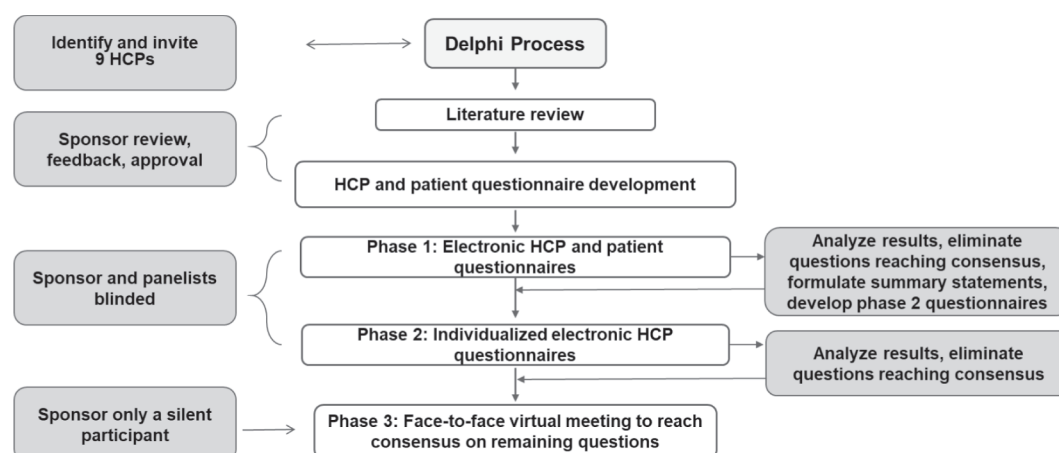


Fig. 1. Delphi methodology and study design. HCP, healthcare provider.

Recommendations

Management of TLS risks and symptoms

Risk factors for TLS

TLS risk classification is based on the expected incidence of clinical TLS, as determined by reports in the literature, and characterized as low (<1% chance), intermediate (~1–5%), or high (>5%) risk [2]. However, the definition of “intermediate risk” for TLS varied among the panelists and depended on the TLS risk stratification system used. Some advocated collapsing risk stratification into “at risk” versus “not at risk,” a binary system that precludes nuanced stratification but greatly simplifies management. This binary system lends itself to a “trigger tool” [20,21] and increases the feasibility of implementing interventions to prevent and manage TLS.

Many individual factors contribute to the overall risk of TLS, including older age, advanced stage of cancer, bulky disease, bone marrow involvement, mediastinal mass, high white blood cell (WBC) count, high lactate dehydrogenase level, dehydration, concurrent medications (other than cancer therapy), renal impairment, hepatomegaly or splenomegaly with tumor involvement, and tumor infiltrating the kidneys. Major risk factors, each of which contributes substantially and independently to the risk of clinical TLS, are bulk of disease, potential for cell lysis (a combination of tumor type, its proliferation rate, its chemosensitivity, and intensity of initial therapy), and presentation (laboratory TLS at presentation, pre-existing AKI). Other comorbidities mentioned as important for TLS risk classification were baseline electrolyte abnormalities (hyperkalemia, hyperphosphatemia, or hyperuricemia); pre-existing cardiac or pulmonary conditions; and ability to receive hyperhydration, defined as > 3000 mL/m²/day.

Renal impairment would include preexisting uremia or hyperuricemia, decreased urine flow, acidic urine, dehydration, oliguria, anuria, renal insufficiency, and AKI. The contribution of kidney function to TLS risk is high for patients receiving renal replacement therapy or dialysis and those with glomerular filtration rate (GFR) < 30 mL/min/1.73 m². The contribution to TLS risk is intermediate for those with GFR > 30–90 mL/min/1.73 m² and low for those with GFR > 90 mL/min/1.73 m². However, the panel noted that their threshold for pediatric patients is lower; that is, children with lower GFR may be classified as higher risk than an adult with the same GFR.

Bulk of disease should be determined by the total tumor mass including all sites of disease (bone marrow, lymph nodes, liver, spleen, and/or other) [22]. Panelists agreed that tumor masses ≤ 400, >400–700, and > 700 g would respectively be considered at low, intermediate, and high risk of TLS. Obstacles to determining the bulk of disease include radiologists not routinely calculating/reporting tumor volume on imaging and difficulty of determining tumor volume in many cases.

TLS risks associated with different malignancies are shown in Table 1. Panelists did not reach consensus on TLS risk in chronic lymphocytic leukemia (CLL) treated with venetoclax and all lymph nodes < 5 cm and absolute lymphocyte count < 25K/μL, with 50% choosing low risk and 50% intermediate risk. Of note, the venetoclax package insert states that this patient scenario would be low risk but recommends similar prophylaxis and blood chemistry monitoring frequency for patients whose tumor burden is low or medium risk [23].

Concern for TLS risk associated with specific anticancer agents is shown in Table 2. The effectiveness of venetoclax in inducing apoptosis of CLL cells confers an increased risk of TLS. Panelists were in favor of using dose-escalation strategies for chemotherapeutic agents (for those with supporting data) to reduce the risk of TLS and noted that this strategy has considerably attenuated the risk of TLS with venetoclax treatment [24–26]. Of note, the incidence of clinical or laboratory TLS was 4.2% in 118 low-risk patients with CLL treated with venetoclax, 10.7% in 94 intermediate-risk patients, and 12.0% in 83 high-risk patients in a large, real-world series of patients [27]; TLS risk was defined

Table 1

TLS risk in various malignancies.

	Patient risk of TLS (consensus)		
	Low	Intermediate	High
<i>Leukemias</i>			
ALL		X	
ALL with WBC < 25K/μL		X	
ALL with WBC 25K–99K/μL		X	
ALL with WBC ≥ 100K/μL			X
AML (all types)		X	
AML with WBC < 25K/μL		X	
AML with WBC 25K–99K/μL			X
AML with WBC ≥ 100K/μL			X
Burkitt leukemia			X
CLL (all types)		X	
CLL receiving only alkylating agents	X		
CLL receiving targeted and/or biologic therapies		X	
CLL treated with venetoclax		X	
CLL treated with venetoclax and all lymph nodes < 5 cm and ALC < 25K/μL	No consensus 50.0%)	No consensus 50.0%)	
CLL treated with venetoclax and any lymph node 5 to < 10 cm and ALC ≥ 25K/μL		X	
CLL treated with venetoclax and any lymph node ≥ 10 cm or ALC ≥ 25K/μL and any lymph node ≥ 5 cm			X
CML (all types)		X	
CML excluding blast crisis	X		
CML-BC			X
<i>Lymphomas</i>			
Anaplastic large-cell lymphoma		X	
Burkitt lymphoma			X
Cutaneous T-cell lymphoma	X		
DLBCL (all types)		X	
DLBCL stage 1 or 2		X	
DLBCL stage 3 or 4		X	
DLBCL with LDH WNL		X	
DLBCL with LDH greater than ULN, non-bulky		X	
DLBCL with LDH greater than ULN, bulky			X
Follicular lymphoma	X		
Hodgkin lymphoma	X		
Lymphoblastic lymphoma (all types)		X	
Lymphoblastic lymphoma, early stage with LDH < 2 × ULN		X	
Lymphoblastic lymphoma, advanced stage			X
Mantle cell lymphoma (all types)		X	
Mantle cell lymphoma, blastoid variants		X	
Mantle cell lymphoma, blastoid variants, with LDH greater than ULN, non-bulky		X	
Mantle cell lymphoma, blastoid variants, with LDH greater than ULN, bulky			X
Mantle cell lymphoma, non-blastoid variants		X	
Marginal zone B-cell lymphoma	X		
Peripheral T-cell lymphoma (all types)		X	
Peripheral T-cell lymphoma with LDH greater than ULN, non-bulky		X	
Peripheral T-cell lymphoma with LDH greater than ULN, bulky		X	
Small lymphocytic lymphoma	X		
T-cell lymphoma (all types)		X	
T-cell lymphoma with LDH greater than ULN, non-bulky		X	
T-cell lymphoma with LDH greater than ULN, bulky			X
Transformed lymphoma (all types)		X	
Transformed lymphoma with LDH greater than ULN, non-bulky		X	

(continued on next page)

Table 1 (continued)

	Patient risk of TLS (consensus)		
	Low	Intermediate	High
Transformed lymphoma with LDH greater than ULN, bulky			X
<i>Other malignancies</i>			
Germ-cell tumors		X	
MDS	X		
Myeloma (all types)	X		
Myeloma with extramedullary disease	X		
Myeloma with extramedullary disease and LDH greater than ULN		X	
Myelofibrosis (all types)	X		
Myelofibrosis—intermediate-risk disease	X		
Myelofibrosis—high-risk disease	X		
Neuroblastoma		X	
Plasma cell leukemia		X	
Small cell lung cancer		X	
Solid tumors (excluding neuroblastomas, germ-cell tumors, and small cell lung cancer)	X		

Bold X indicates agreement with the Cairo 2010 [2] risk classification.

ALC, absolute lymphocyte count; ALL, acute lymphoblastic leukemia; AML, acute myeloid leukemia; CLL, chronic lymphocytic leukemia; CML, chronic myeloid leukemia; CML-BC, chronic myeloid leukemia in blast crisis; DLBCL, diffuse large B-cell lymphoma; LDH, lactate dehydrogenase; MDS, myelodysplastic syndrome; TLS, tumor lysis syndrome; ULN, upper limit of normal; WBC, white blood cells; WNL, within normal limits.

Table 2

Degree of concern of TLS risk with chemotherapeutic agents.

	Level of concern for TLS risk (consensus)				
	No concern	Low concern	Some concern	High concern	Most concern
Treatment					
Alemtuzumab			X		
Alvociclib (flavopiridol)		X			
Bortezomib		X			
Brentuximab vedotin		X			
Carfilzomib		X			
CAR T			X		
Clofarabine			X		
Daratumumab			X		
Dasatinib		X			
Dinaciclib		X			
Etoposide			X		
Ibrutinib			X		
Idelalisib			X		
Lenalidomide		X			
Mogamulizumab			X		
Obinutuzumab			X		
Ofatumumab			X		
Oprozomib		X			
Ponatinib		X			
Rituximab			X		
Venetoclax				X	

CAR T, chimeric antigen receptor T cells; TLS, tumor lysis syndrome.

as in the venetoclax package insert. Dose escalation has also been used successfully with venetoclax in children and adults with acute myeloid leukemia [28–31].

Many guidelines do not mention tumor debulking to reduce the risk of TLS, although there are several published trials reporting that this approach reduces the risk. Initial ibrutinib monotherapy prior to venetoclax treatment in patients with CLL led to TLS risk downgrades in 80% [32] and 90% [33] of high-risk patients. TLS risk was also reduced in treatment-naïve patients with CLL who received a 21-day pretreatment with rituximab, obinutuzumab, or obinutuzumab plus ibrutinib,

all prior to treatment with venetoclax [34]. In children with acute lymphoblastic leukemia (ALL), a 1-week prednisone pre-phase has been used for decades to assess treatment response for risk-group assignment, to decrease the risk of TLS, and to allow management of comorbidities prior to starting multi-drug induction chemotherapy [35,36].

Monitoring and prophylaxis for TLS

The panel agreed that monitoring for TLS should begin prior to the first dose of therapy regardless of the TLS risk. A panelist noted that all patients should have baseline laboratory values recorded. For TLS monitoring, the panel considered urine output, potassium, phosphate, uric acid, calcium, creatinine, blood urea nitrogen, and lactate dehydrogenase to be important parameters to monitor for TLS (Table 3). Monitoring sodium, carbon dioxide, chloride, and electrocardiogram (ECG) were considered somewhat important, with urine pH being not important. Outside of the questionnaire, monitoring bicarbonate was considered somewhat important. The importance of ECG or continuous cardiac monitoring depends on tumor bulk, degree of TLS risk, specific anticancer drug utilized, and most importantly, the presence of hyperkalemia or hypocalcemia. Panelists did not take measurements at certain times of day, but some took diurnal variation into account.

Panelists used cancer type, cancer stage, specific oncologic therapy, and renal function in their determination of how often to monitor for TLS. Panelists would consult a nephrologist for patients with clinical TLS and a potential need for dialysis. In determining whether a patient should be managed on an inpatient or outpatient basis, important factors were: the degree of risk of TLS, whether the patient had established TLS, cancer type, cancer stage, specific oncologic therapy, acute oncologic emergency, uric acid level, phosphorus level, potassium level, oliguria, and renal function. The presence of a mediastinal mass, institutional protocol, financial reasons, and proximity of the patient's home to the medical center were important to somewhat important in this decision.

Monitoring for TLS in the inpatient setting. In the inpatient setting, the panel agreed that patients at low risk of TLS would be monitored once a day. No consensus was obtained for the timing of monitoring of intermediate-risk patients, with 62.5% choosing every 12 h (or twice daily). The panelists noted that intermediate risk is a difficult category, because some of these patients are treated in the outpatient setting; this population is very heterogeneous in that some require more frequent monitoring whereas others do not. Of note, guidelines recommend laboratory monitoring of intermediate-risk patients every 8–12 h [4]. Panelists noted that it is neither feasible nor routine clinical practice to admit all intermediate-risk adult patients for TLS monitoring.

Patients at high risk of TLS would be monitored every 6 h in the inpatient setting, with those with laboratory or clinical TLS monitored every 4 h. Some panelists who care primarily for adult patients noted that monitoring intervals of < 8 h were not feasible at their institutions because there would not be sufficient time to receive laboratory results, adjust the intervention(s), and accurately assess the effect of the intervention(s). In contrast, pediatric oncologists routinely monitor their patients every 4–6 h. Panelists noted that any patient with hyperphosphatemia or hyperkalemia would need more frequent electrolyte monitoring.

Monitoring for TLS in the outpatient setting. In the outpatient setting on the day of chemotherapy treatment, panelists agreed that patients at low risk of TLS would be monitored once daily, in agreement with guidelines [4]. There was no consensus on how often to monitor intermediate-risk patients. Patients at high risk of TLS or those with laboratory or clinical TLS were excluded from this scenario, because they would be treated in the inpatient setting. Several panelists noted that it would be difficult to obtain laboratory results more than once or twice a day for outpatients.

In the outpatient setting after chemotherapy treatment, patients at a low risk of TLS would be monitored once a day. Again, no consensus was

Table 3
Consensus-based recommendations for TLS prophylaxis.

Parameter	TLS risk			Comments
	Low	Intermediate	High	
Laboratory testing	Outpatient; once daily	- Could be inpatient or outpatient - No consensus, but 62.5% would monitor every 12 h/twice daily	Inpatient; every 6 h	- Monitor urine output, potassium, phosphate, uric acid, calcium, creatinine, BUN, LDH - Patients should be monitored as long as they are at risk of TLS
Hydration	Oral fluids for a 2–3 L/m² daily volume starting prior to chemotherapy that might put them at risk of TLS until the end of the day following chemotherapy treatment	Continuous IV infusion of 2–3 L/m² daily starting prior to chemotherapy, usually upon admission, until the day following chemotherapy treatment	Continuous IV infusion of 2–3 L/m² daily starting prior to chemotherapy, usually upon admission, until the day following chemotherapy treatment	- Monitor hydration status every 6–8 h - Target urine output of 80–100 mL/m²/h - For those in the inpatient setting, a continuous IV infusion is recommended, whereas those in the outpatient setting would be instructed to drink oral fluids - Diuretics (furosemide) for patients whose urine output is low after hyperhydration and in patients with fluid overload (e.g., peripheral edema)
Hyperkalemia	Avoid foods high in potassium	Avoid foods high in potassium	- Avoid foods high in potassium - Continuous cardiac monitoring in patients with potassium ≥ 6 mmol/L	
Hypocalcemia	No prophylaxis	No prophylaxis	No prophylaxis	
Hyperphosphatemia	Low phosphorus diet	Hydration	Hydration	
Hyperuricemia	- Laboratory monitoring - Oral hydration - Allopurinol (febuxostat for those hypersensitive to allopurinol)	- Laboratory monitoring - Oral hydration - Continuous IV hydration - Allopurinol (in adult and pediatric patients; febuxostat for those hypersensitive to allopurinol)	- Laboratory monitoring - Oral hydration - Continuous IV hydration - Allopurinol (in adults; febuxostat for those hypersensitive to allopurinol) - Rasburicase (routinely in children, in selected cases only with adults [serum uric acid is > 7.5 mg/dL (or above ULN for age and sex) in patients with established clinical or laboratory TLS, high risk of TLS, serum creatinine rise of ≥ 0.3 mg/dL (or $> 50\%$ increase from baseline), bulky disease, and high WBC count (for patients with acute leukemias)])	- Initiate allopurinol (febuxostat for those hypersensitive to allopurinol) treatment 2–3 days prior to chemotherapy but do not delay initiation of chemotherapy while waiting for uric acid to decrease - Allopurinol treatment to continue depending on TLS risk - Rasburicase may be considered concurrently with allopurinol in adult or pediatric patients - Note: allopurinol administration prior to and then concurrent with rasburicase treatment (if rasburicase is required) is standard in most adult institutions and less common in pediatric care facilities - Pediatric patients with large tumor burden and chemotherapy-sensitive cancer such as Burkitt leukemia/lymphoma, T-cell ALL, and monoblastic leukemia commonly receive rasburicase as prophylaxis with or without allopurinol - For those unable to take oral allopurinol, rasburicase is preferred over IV allopurinol due to cost - No urine alkalization

BUN, blood urea nitrogen; IV, intravenous; LDH, lactate dehydrogenase; TLS, tumor lysis syndrome.

reached on monitoring frequency for those at intermediate risk, with 57% recommending every 12 h or twice daily. As mentioned previously, patients at high risk of TLS or those with laboratory or clinical TLS would be treated in the inpatient setting.

Panelists agreed that patients should be monitored for TLS if they are at risk; no longer at risk typically corresponds with debulking and serial laboratory results demonstrating no evidence of TLS. Barriers to outpatient monitoring for TLS include personnel, resources, time to receive laboratory results, and patient accessibility.

Hydration

Aggressive hydration is a cornerstone of TLS prophylaxis and management, as it allows for dilution of potentially harmful levels of solutes from tumor lysis. Panelists would instruct a low-risk patient to drink oral

fluids for a volume of 2–3 L/m² daily starting prior to any therapy that might put them at risk of TLS until the end of the day following anti-cancer treatment completion (Table 3). A few panelists noted that they may advise oral hydration for 1–2 days following therapy. Intermediate- and high-risk patients would receive a continuous intravenous (IV) infusion of 2–3 L/m² daily, starting prior to therapy (usually upon admission) on the day of therapy and the following day. For patients with clinical or laboratory TLS, a saline bolus followed by continuous IV infusion is recommended (Table 4).

Hydration status would be monitored every 6–8 h for inpatients, with a recommended target urine output of 80–100 mL/m²/h in agreement with guidelines [3] and monitoring for symptoms of volume overload (peripheral edema, weight, respiratory status). Diuretics would be used in patients at risk of TLS whose urine output is low but only after

Table 4

Consensus-based recommendations for management of laboratory or clinical TLS.

Parameter	Recommendation
Laboratory testing	• Inpatient; every 4–6 h
Hydration	• Saline bolus followed by continuous IV infusion 2–3 L/m²/day; isotonic fluids preferred • Avoid fluids containing potassium, calcium, or phosphate (e.g., lactated Ringer's solution) • Monitor hydration status every 6–8 h • Target urine output of 80–100 mL/m²/h • Diuretics may be used concomitantly with IV hydration for management of volume overload and maintenance of adequate urine output to match input
Hyperkalemia	• Continuous cardiac monitoring in patients with potassium ≥ 6 mmol/L or with AKI and TLS • Arrhythmia and acute cardiotoxicity managed with calcium gluconate infusion and ECG monitoring • If any ECG changes, give IV calcium gluconate 1 g IVPB $\times 1$ over 30 min • Treat hyperkalemia in established laboratory TLS with sodium zirconium cyclosilicate 10 g three times daily, sodium polystyrene sulfonate 15 g every 6 h until potassium is normal, or patiomer 8.4 g once daily • Treat patients who fail to respond to oral potassium-lowering agents and those with symptomatic hyperkalemia with rapid-acting insulin, glucose infusion, and sodium bicarbonate • For example, 50% dextrose 25 g IV followed by insulin lispro 0.1 U/kg (10 U maximum dose) • If patient has concomitant metabolic acidosis, consider sodium bicarbonate 50 mEq IV • For refractory or severe hyperkalemia, consider albuterol 10 mg inhalation as needed • Defer to nephrologist or ICU physician for decision to initiate renal replacement therapy or dialysis
Hypocalcemia	• No treatment of asymptomatic hypocalcemia • Pharmacological treatment for symptomatic hypocalcemia initiated when calcium is ≤ 7 mg/dL or ionized calcium is ≤ 0.8 mmol/L • Calcium gluconate IV for clinical TLS at lowest dose possible to relieve symptoms and avoid precipitation of calcium phosphate; 1000 mg (adults) or 100 mg/kg up to 1000 mg (children) IV infusion over 30 min
Hyperphosphatemia	• Dietary phosphate restriction • Increased IV hydration • Elimination of phosphate from IV solutions • Use phosphate binders: calcium carbonate 1000 mg every 6 h for adults, 400 mg every 8 h for children weighing < 20 kg; sevelamer 0.8 or 1.6 g three times daily with meals; lanthanum carbonate 1500 mg daily in divided doses • Dialysis or hemofiltration only if severe hyperphosphatemia is accompanied by other issues that warrant dialysis (life-threatening hyperkalemia, symptomatic fluid overload)
Hyperuricemia	• Laboratory monitoring • Oral hydration • Increased IV hydration • Rasburicase (routinely in pediatric patients with chemotherapy sensitive disease, such as Burkitt leukemia, Burkitt lymphoma with large tumor burden, T-ALL and monoblastic leukemia), if not contraindicated. Allopurinol can be added after rasburicase treatment. • Allopurinol (febuxostat for those hypersensitive to allopurinol). Note: allopurinol administration prior to and then concurrent with rasburicase treatment (if rasburicase is required) is standard in most adult institutions and less common in pediatric care facilities. • No urine alkalinization

AKI, acute kidney injury; ECG, electrocardiogram; ICU, intensive care unit; IV, intravenous; IVPB, intravenous piggyback; TLS, tumor lysis syndrome.

hyperhydration and in patients with fluid overload, with furosemide being the diuretic of choice.

Management of acute effects of TLS

Hyperkalemia/potassium management

Hyperkalemia is defined as a serum potassium level above the upper limit of normal (ULN), which is usually > 5.0 – 5.5 mmol/L, and can lead to cardiac abnormalities such as dysrhythmia that can result in sudden death [37]. In regard to TLS, the Cairo-Bishop guidelines define hyperkalemia as potassium levels ≥ 6 mmol/L or a 25% increase from baseline [8]. The Howard guidelines posit that a 25% increase from baseline may not be clinically important unless the baseline value is above the normal range [4]. Panelists noted that the serum potassium level is age-dependent and that an increase of 25% from baseline should be a concern. For prophylaxis (Table 3), panelists recommended that patients at any risk level of TLS avoid foods high in potassium.

Because of the risk of sudden death even with asymptomatic hyperkalemia, pharmacological intervention should be considered at serum potassium levels ≥ 5.5 mmol/L. Continuous cardiac monitoring for hyperkalemia should be used in patients with a potassium level ≥ 6 mmol/L and in those with AKI and TLS (Table 4). Arrhythmias and acute cardiotoxicity due to hyperkalemia should be managed via calcium gluconate infusion and/or insulin and glucose and ECG monitoring, in agreement with prior guidelines [3,19]. Arrhythmia is of great concern, as a large US database review identified arrhythmia among 24% of patients hospitalized with TLS, and TLS patients with arrhythmia had significantly higher in-hospital mortality rates compared with TLS patients without arrhythmia (32% vs 21%; $p < 0.001$) [38].

Patients with established laboratory TLS and hyperkalemia may initially be treated with oral potassium-lowering agents (sodium zirconium cyclosilicate, sodium polystyrene sulfonate, or patiomer; the specific agent often depends on institutional formulary) in addition to hydration and (as needed) diuresis. Hydration is a crucial component of care, and more aggressive fluid rates should be considered in patients with hyperkalemia. Patients not responding to oral potassium-lowering agents and those with symptomatic hyperkalemia would also be treated with rapid-acting insulin and a glucose infusion, as in prior guidelines [3,19]. Sodium bicarbonate may also be used, though it is generally only beneficial in patients with concomitant metabolic acidosis and not part of managing any other aspects of TLS. The decision to initiate renal replacement therapy or dialysis for hyperkalemia would be deferred to the nephrologist or intensive care unit physician.

Management of hypocalcemia should be especially aggressive in patients who also have hyperkalemia. Hypocalcemia is treated with calcium gluconate only when symptomatic with paresthesias, tetany, or other symptoms because increasing the calcium level increases the risk of calcium phosphate crystal precipitation in the kidneys and cardiac conduction system. However, in patients with hyperkalemia, the risk of cardiac dysrhythmias and sudden cardiac death outweighs the risks posed by increased calcium phosphate crystal precipitation, as discussed below.

Hypocalcemia/calcium management

The Cairo-Bishop guidelines define hypocalcemia as calcium levels ≤ 1.75 mmol/L (≤ 7 mg/dL) or a 25% decrease from baseline [8]. Again, the Howard guidelines do not consider the 25% decrease from baseline to be clinically relevant unless the baseline is already below the normal range but confirm that symptomatic hypocalcemia constitutes clinical TLS [4]. Extreme hypocalcemia can lead to cardiac arrhythmia, tetany, muscle cramps, and hypotension [1]. The panel would measure both ionized and total calcium. There was no consensus on how often to measure serum calcium, although it was noted that the frequency of blood draws would be dictated by electrolytes other than calcium, and that all relevant electrolytes should be evaluated at each time point.

The panelists recommended against the use of prophylaxis for

hypocalcemia for all risk levels of TLS (Table 3), noting that calcium should not be given in the absence of hypocalcemia, because the potential to precipitate as calcium phosphate exists, which can lead to AKI or cardiac dysrhythmia (if it precipitates in the cardiac conduction system).

Panelists did not recommend treatment of asymptomatic hypocalcemia (Table 4), consistent with guidelines [3,4,19]. Calcium gluconate infusion would be used to treat symptomatic hypocalcemia when calcium is ≤ 7 mg/dL or ionized calcium is ≤ 0.8 mmol/L (≤ 3.2 mg/dL), repeated every hour until symptoms resolve or calcium levels normalize. Regarding management of patients with concomitant hyperphosphatemia and symptomatic hypocalcemia, panelists recommended using the lowest dose of calcium sufficient to relieve symptoms.

Management of TLS to address other electrolyte abnormalities and to prevent long-term renal effects

Hyperphosphatemia/phosphate management

The Cairo-Bishop definition of hyperphosphatemia is phosphorus levels ≥ 2.1 mmol/L (≥ 6.5 mg/dL) in children and ≥ 1.45 mmol/L (≥ 4.5 mg/dL) in adults, or a 25% increase from baseline [8]; the Howard guidelines only consider the 25% increase from baseline to be of clinical value when the baseline value is higher than normal [4].

Hyperphosphatemia is of concern, because it can lead to precipitation of calcium phosphate crystals, which can cause AKI, cardiac arrhythmias, and worsened hypocalcemia [1]. An increase in serum phosphate levels is associated with development of AKI in adults and children with TLS [39–42].

Panelists recommended a low phosphorus diet to prevent hyperphosphatemia (Table 3), in agreement with guidelines [4]. Hydration was also recommended for those at intermediate to high risk of TLS.

Treatment recommendations for hyperphosphatemia include dietary phosphate restriction, increased hydration, elimination of phosphate from IV solutions, and use of phosphate binders (Table 4), recommendations consistent with guidelines [3]. Panelists recommend the phosphate binders calcium carbonate, sevelamer, or lanthanum carbonate. Of note, lanthanum can interfere with abdominal X-rays and appear radiopaque [43], whereas calcium carbonate can increase calcium levels and the risk of calcium phosphate precipitation and crystal formation. The panelists do not use dialysis or hemofiltration for severe hyperphosphatemia unless accompanied by other clinical or laboratory issues that warrant dialysis such as life-threatening hyperkalemia or symptomatic fluid overload.

Hyperuricemia management

The Cairo-Bishop [8] definition of hyperuricemia is uric acid ≥ 476 μ mol/L (≥ 8 mg/dL) or 25% increase from baseline. The Howard guidelines do not include the 25% increase from baseline in adults and define hyperuricemia in children as above the ULN for age [4].

Hyperuricemia is a concern because the accumulation of uric acid may lead to crystallization and/or coprecipitation with calcium phosphate with subsequent damage of renal tubules [4]. Allopurinol is a xanthine oxidase inhibitor that blocks the conversion of hypoxanthine and xanthine to uric acid, although it does not affect the excretion of existing uric acid [44]. Because xanthine is converted to uric acid, blockage of its conversion by allopurinol leads to accumulation of xanthine. As xanthine is far less soluble than uric acid, xanthine crystals form at a higher rate than uric acid crystals; thus, xanthine nephropathy can lead to AKI in patients with normal uric acid (xanthine is measured only in research labs) and cannot be managed because no treatment option exists that can eliminate xanthine. Panelists noted historical case reports of xanthine nephropathy with the use of allopurinol [45–47], but the role of xanthine in TLS has not been documented with contemporary treatment. Most panelists are not concerned by this issue. Febuxostat is also a xanthine oxidase inhibitor that, in the experience of the panel, is not used as often as allopurinol because febuxostat lacks definitive

evidence that it can reduce serum uric acid levels faster than allopurinol at equipotent doses [48] and is more expensive. In patients with gout and cardiovascular disease, higher all-cause and cardiovascular mortality were seen with febuxostat compared with allopurinol over prolonged periods of time [49]. However, febuxostat can be used in patients with hypersensitivity to allopurinol [1]. Neither allopurinol nor febuxostat reduce uric acid that formed prior to treatment, so effects are delayed 24–72 h [50]. Another treatment option is rasburicase, a recombinant urate oxidase that converts uric acid to allantoin, which is more soluble and more readily excreted than uric acid [51]. Rasburicase is contraindicated in those with glucose-6-phosphate dehydrogenase deficiency or a history of anaphylaxis, severe hypersensitivity, hemolysis, or methemoglobinemia with prior rasburicase treatment [52].

Panelists recommend measurement of baseline uric acid in patients at all risk levels of TLS and in those with impaired renal function (Table 3). Different strategies are often used for pediatric versus adult patients. Prophylactic measures for those at any risk level of TLS include laboratory monitoring, oral hydration, IV hydration (except for those at low risk), and allopurinol. This differs from other guidelines that recommend rasburicase for prophylaxis in patients at high risk of TLS [2–4,19]. Patients with laboratory or clinical TLS should be managed with laboratory monitoring, oral hydration, IV hydration, allopurinol (in adults), and rasburicase (generally used in pediatric patients with hyperuricemia, large tumor burden, and chemotherapy-sensitive cancer, especially Burkitt leukemia/lymphoma, T-cell ALL, and monoclonal leukemia) (Table 4). Several panelists noted that there are institution-based limitations on the use of rasburicase in adults because of cost issues. As rasburicase has a rapid onset of action [53], one panelist noted that it can be given as needed if uric acid rises or fails to decrease in a timely manner despite prophylaxis with fluids and allopurinol. However, another panelist suggested that rapid reduction of uric acid by rasburicase can prevent crystallization and/or co-precipitation with calcium phosphate and enhance phosphate excretion.

The panelists recommend that allopurinol treatment be initiated 2–3 days prior to therapy when possible but not to further delay the initiation of chemotherapy while waiting for the uric acid levels to decrease, with allopurinol treatment continuing until the risk of TLS has decreased. When more urgent anticancer therapy is indicated in patients with rapidly progressive disease, panelists would not delay therapy for 2–3 days for allopurinol treatment and would use rasburicase instead when indicated. An area of active discussion was the timing of allopurinol initiation. Most adult patients receive allopurinol prior to therapy, and if rasburicase is required, will receive both concurrently. Although this strategy is common, it has not been studied in clinical trials. Dosing for allopurinol did not reach consensus, with 55.6% agreeing with the modal answer of 600–800 mg/day for adult patients; Jones et al. recommend 200–400 mg/m²/day in 1–3 divided doses for a maximum of 800 mg/day in adults [19]. Some panelists give a loading dose of 600–900 mg and then a lower maintenance dose; others divide a 600 mg dose in higher risk patients into 300 mg twice daily. Alternatively, weight-directed dosing is 10 mg/kg/day divided every 8 h (maximum 800 mg/day) and body surface area-directed dosing 50 to 100 mg/m²/dose (maximum 300 mg/m²/day). Because allopurinol is excreted by the kidneys, a dose reduction of 50% is recommended in patients with renal insufficiency. However, this reduction generally applies for the treatment of gout, and it is often not applied to patients with TLS [54]. Dosing for pediatric patients is shown in Table 3. Most patients would benefit from aggressive hydration and regular uric acid monitoring regardless of allopurinol administration.

Regardless of the risk at presentation, clinicians should reassess patients for TLS risk at regular intervals based on laboratory values, urine output, and clinical conditions. In those who present at low or intermediate risk of TLS but later develop hyperuricemia or laboratory TLS, management should be as for patients at high risk, because hyperhydration and allopurinol were insufficient for such patients. Treatment with rasburicase should be started when serum uric acid is > 7.5 mg/dL

(or above ULN for age and sex) in patients with established clinical or laboratory TLS, high risk of TLS, serum creatinine rise of ≥ 0.3 mg/dL (or $> 50\%$ increase from baseline), bulky disease, and high WBC count (for patients with acute leukemias).

Despite the 0.2 mg/kg/dose once daily for up to 5 days on the US product label for rasburicase [52], several reports used lower doses and shorter durations of treatment. Indeed, single-dose rasburicase is frequently used at 0.15 mg/kg or as a fixed dose of 3–7.5 mg, with repeat doses given based on subsequent uric acid levels. A meta-analysis of 19 studies found that a single-dose regimen was often effective [55]. Response rates ranged from 68% (4.5 mg fixed dose) to 85–99% (7.5 mg fixed dose or the weight-based 0.15 mg/kg). Of note, six of the analyzed studies included < 20 subjects. The panelists recommend a capped 3 mg dose of rasburicase for adults. When a single dose is used, close monitoring of uric acid is warranted to determine if subsequent dosing is needed, because 3 mg is not sufficient for all patients. One panelist noted that patients weighing ≥ 100 kg treated at their institution receive 4.5 mg for the first dose; another uses 6 mg. Patients with uric acid levels > 15 mg/dL may also require higher initial doses. In this setting, one panelist reported their institutional guideline uses a 6-mg dose and recommends additional doses if uric acid levels continue to rise despite the initial dose [56]. Pediatric dosing is often weight-based, at the Food and Drug Administration (FDA)-approved 0.2 mg/kg daily dose [52], but this is not uniform among the panelists' institutions. A formal non-inferiority study is needed to definitively address rasburicase dosing requirements and to compare them with the FDA-approved dose. Rasburicase should be infused over 30 min, and the dose should be rounded down to the nearest vial size.

Based on the experience of the panelists and consistent with results from several studies [57–60], several benefits are observed with the use of rasburicase. A retrospective analysis of rasburicase versus allopurinol found significantly greater reductions in uric acid with rasburicase as well as significantly shorter hospital and intensive care unit stays and lower total costs per patient, although it is unclear whether the hospital admittances and lengths of stays were directly due to TLS [58]. Reductions in uric acid levels, improvements in GFR [59], and improvements in renal function as measured by increased creatinine clearance [60] have been observed with rasburicase treatment. In a study of pediatric B-cell lymphoma, French patients, who all received non-recombinant urate oxidase, had significantly lower TLS rates (9% vs. 26%) and dialysis rates (3% vs 15%) than North American patients who had no access to urate oxidase [57]. Perhaps because of these promising reports, there have been no randomized controlled trials to definitively prove the renal-protective effect of rasburicase.

Alkalinization of urine

Panelists recommend against alkalinization of urine (Tables 3, 4) in accordance with guidelines [3,4,19], as urine alkalinization may promote calcium phosphate precipitation [4].

Limitations to this study

The modified Delphi process used here is a well-established method to achieve consensus of experts, even when individual opinions about specific aspects may be controversial, with strong minority opinions. It also effectively captures current opinions and current practices. However, evidence-based clinical care must not be based solely on consensus opinions but also on published peer-reviewed data. For example, many panelists routinely use reduced doses of rasburicase that have not been tested against approved doses in randomized, controlled trials, so consensus opinion must be put in the context of established regulatory approvals based on clinical trials. Where no clinical trials exist, the Delphi process can be effective and compelling, as in the case of management of IV fluids, hyperkalemia, hyperphosphatemia, and hypocalcemia in people at risk of TLS. Notably, pediatric and adult oncologists may have varying approaches to TLS, because they are treating different

diseases, obtaining different responses in the same disease, or using different therapeutics. Treatment approaches also differ between academic and community oncology centers. Finally, the Delphi process may not effectively incorporate innovations in care, such as the use of a binary risk classification system for TLS (“not at risk” versus “at risk”) that greatly simplifies risk stratification and makes effective management of TLS more structured and practical.

Perspectives

This modified Delphi panel reached consensus on many aspects of risk assessment, monitoring, prevention, and management of TLS. Several panelists expressed key differences from published guidelines based on their clinical experience and differences in the monitoring, prophylaxis, and treatment of TLS between adults and children with cancer. With the availability of xanthine oxidase inhibitors and rasburicase, uric acid levels can be well-controlled; however, clinicians must remain aware of other dangerous but treatable electrolyte abnormalities in TLS.

CRediT authorship contribution statement

Anthony J. Perissinotti: Data curation, Writing – review & editing. **Michael R. Bishop:** Data curation, Writing – review & editing. **Joseph Bubalo:** Data curation, Writing – review & editing. **Mark B. Geyer:** Data curation, Writing – review & editing. **Amy Goodrich:** Data curation, Writing – review & editing. **Scott C. Howard:** Conceptualization, Data curation, Writing – review & editing. **Julianna Kula:** Data curation, Writing – review & editing. **Sreedhar Mandayam:** Data curation, Writing – review & editing. **Mitchell S. Cairo:** Data curation, Writing – review & editing. **Ching-Hon Pui:** Data curation, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: AJP reports consulting fees from AbbVie, BeiGene, Pfizer, Sanofi-Aventis, and Servier. MRB serves as a member on an advisory board or as a consultant for Autolus Therapeutics, BMS/JUNO Therapeutics, Chimeric Therapeutics, CRISPR Therapeutics, IN8bio, Incyte, Iovance Biotherapeutics, KITE/Gilead, Novartis, PhysiQ, and Sana Biotechnology and is on the speakers' bureau for ADC Therapeutics, Agios, BMS, Kite/Gilead, Incyte, Sanofi, and Servier. JB reports consulting fees from Sanofi-Aventis. MBG reports research support from Actinium Pharmaceuticals, Amgen, and Sanofi and has served on an advisory board for Sanofi and as a consultant for Novartis. AG reports consulting fees from Sanofi-Aventis. SCH has served as a speaker for Jazz Pharma and Sanofi, has received grant funding from Jazz Pharma, and has served as a consultant for Servier. JK reports consulting fees from Sanofi-Aventis. SM reports grant funding from Horizon; service on advisory boards for Calliditas, Chinook, Trave, and US Renal Care; and service on speaker's bureaus for Bayer and Calliditas. MSC serves on the speaker's bureau and as a consultant for Sanofi. C-HP reports support by grant CA21765 from the National Cancer Institute and American Lebanese Syrian Associated Charities, and honorarium from Amgen and Novartis.

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